Volumes from segmentation of T1 MRI scans of the brain - Insight 46 Phase 1

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Document details

Categories of variables	Volumes from segmentation of T1 MRI scans of the brain
Summary of work undertaken	1. Source data transferred to Stata file. 2. See references below for methods. 3. Units are cm3 4. Whole brain volumes from Brain MAPS, QC'd and edited. 5. Ventricle (manual segmentation) 6. Left and right hippocampus (STEPS with hippocampal tail included, QC reviewed and edited if necessary) From Statistical Parametric Mapping software (SPM): 1. Total intracranial (TIV) 2. Grey matter (GM) 3. White matter (WM) 4. Cerebrospinal fluid (CSF)
Names of source variables	brain_vol, vent_vol, hippol_vol, hippor_vol, spm_tiv_vol, spm_gm_vol, spm_wm_vol, spm_csf_vol
Output variables	brain_vol_i46p1, vent_vol_i46p1, hippol_vol_i46p1, hippor_vol_i46p1, spm_tiv_vol_i46p1, spm_gm_vol_i46p1, spm_wm_vol_i46p1, spm_csf_vol_i46p1
Papers using these variables	1. Lane et al. (2019). Associations between blood pressure across adulthood and late-life brain structure and pathology in the neuroscience substudy of the 1946 British birth cohort (Insight 46): an epidemiological study. The Lancet Neurology. 2. Liu et al. (in press) Cognition at age 70: life course predictors and associations with brain Pathologies. Neurology

References

Brain MAPS: Leung, K. K., Barnes, J., Modat, M., Ridgway, G. R., Bartlett, J. W., Fox, N. C., ... & Alzheimer's Disease Neuroimaging Initiative. (2011). Brain MAPS: an automated, accurate and robust brain extraction technique using a template library. Neuroimage, 55(3), 1091-1108.

Hippocampal STEPS: Cardoso, M. J., Leung, K., Modat, M., Keihaninejad, S., Cash, D., Barnes, J., ... & Alzheimer's Disease Neuroimaging Initiative. (2013). STEPS: Similarity and Truth Estimation for Propagated Segmentations and its application to hippocampal segmentation and brain parcelation. Medical image analysis, 17(6), 671-684.

Variables in this document

8 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Brain volumes [60331] › Brain volumes - Cross-sectional [603311]		
brain_vol_i46p1	Total brain volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
hippol_vol_i46p1	Left hippocampal volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
hippor_vol_i46p1	Right hippocampal volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
spm_csf_vol_i46p1	Cerebro-spinal fluid volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
spm_gm_vol_i46p1	Grey matter volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
spm_tiv_vol_i46p1	Total intracranial volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
spm_wm_vol_i46p1	White matter volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field
vent_vol_i46p1	Ventricle volume (cm3) - Cross-sectional - Insight46 Phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan